Master's Programme in Data Science

Detection of Anomalies from Gamma Radiation Measurement Data

Yee Hui Min, Joonas Honkavaara, Kim Rosqvist, Ofia Rahman Alif, Mahfuzur Rahman

June 30, 2026

University of Helsinki
Faculty of Science

P. O. Box 68 (Pietari Kalmin katu 5)

00014 University of Helsinki

HELSINGIN YLIOPISTO — HELSINGFORS UNIVERSITET — UNIVERSITY OF HELSINKI

Tiedekunta — Fakultet — Faculty	Koulutusohjelma — Utbildningsprogram — Degree programme
Faculty of Science	Master's Programme in Data Science

Tekijä — Författare — Author

Yee Hui Min, Joonas Honkavaara, Kim Rosqvist, Ofia Rahman Alif, Mahfuzur Rahman

Työn nimi — Arbetets titel — Title

Detection of Anomalies from Gamma Radiation Measurement Data

Työn laji — Arbetets art — Level	Aika — Datum — Month and year	Sivumäärä — Sidantal — Number of pages
Report	June 30, 2026	17

Tiivistelmä — Referat — Abstract

The Radiation and Nuclear Safety Authority of Finland (STUK) collects tens of thousands of environmental samples each year, covering materials such as fish, soil, water, and milk, and manually enters the radioactivity measurements into a database. This manual process introduces the risk of entry errors and unusual values from atypical samples, making reliable anomaly detection important but difficult to perform at scale.

This project built an automated anomaly detection system to support that process. The work involved cleaning three linked datasets, then running clustering methods including autoencoder-based clustering and HDBSCAN to surface potential outliers. Flagged samples were submitted to STUK experts for manual review, and those confirmed labels were used to train Local Outlier Factor models for two tasks: one for preprocessing metadata and one for radioactive measurement results.

Both models classify incoming samples as typical, atypical, or erroneous. The preprocessing model reached 93.16% overall accuracy, with an F1-score of 96.21% on normal samples and 64.52% on anomalies. The measurements model reached 92.97% overall accuracy, with an F1-score of 96.10% on normal samples and 64.50% on anomalies, though performance on anomalies remained harder to improve given the high variability across sample types and nuclide combinations.

The deliverables include Jupyter notebooks for integration into STUK's existing pipeline and two Streamlit web applications for analysts who prefer not to run the code directly.

Avainsanat — Nyckelord — Keywords

anomaly detection, clustering, classification model

Säilytyspaikka — Förvaringsställe — Where deposited

University of Helsinki

Muita tietoja — Övriga uppgifter — Additional information

Contents

Appendix C Clusters of SampleClass

1. Introduction

The Radiation and Nuclear Safety Authority of Finland (STUK) is responsible for monitoring environmental radioactivity in Finland, collecting tens of thousands of samples each year. These samples include a wide range of environmental materials such as fish, soil, milk, and others. The measurements gathered from these samples form a dataset used to assess radiation safety in the environment. Because of this, ensuring accurate and reliable data is critical.

The measurement results are manually entered into a database. This process is prone to human error, and it can also include unusual values due to rare or atypical samples. Identifying such anomalies is an important but demanding task. Manual inspection is slow and difficult given the growing volume of data.

The goal of this project is to automate the detection of anomalies in STUK's measurement database. The approach taken in this project begins with data cleaning and exploratory analysis to better understand the dataset. Clustering techniques are then applied to identify patterns and highlight potential outliers. These findings are reviewed and validated by experts at STUK, after which machine learning models are trained to predict anomalies in new, unseen data.

The main output of the project consists of two components. First, a user interface that allows users to input data and visualize insights, making the system accessible as a standalone tool. Second, raw code scripts that can be integrated into STUK's existing data pipeline and adapted to their specific needs. This report presents the work step by step. It begins with the technical approach, including the tools used and the processes for data cleaning, exploration, clustering, and model development. This is followed by a description of the deliverables, including results, pipeline design, and user interface implementation. Finally, the report discusses challenges encountered during the project, potential future improvements, and concluding remarks.

The project delivered two working prototypes, one for preprocessing metadata and one for radioactive measurement results, both using a Local Outlier Factor model. Each model classifies incoming samples as typical, atypical, or erroneous. The final deliverables include Jupyter notebooks for integration into STUK's existing workflows and Streamlit web applications for interactive use.

2. Technical Approach

2.1 Tools and Technological Stack

The project was implemented in Python, using Pandas and NumPy for data handling, Matplotlib and Seaborn for visualisation, and scikit-learn for clustering and anomaly detection. Jupyter Notebooks supported iterative exploration and documentation throughout the analysis phases. The interactive prototype interfaces were built with Streamlit. Version control and task coordination were handled via Git and GitHub, team communication via Discord and Microsoft Teams, and documentation via Google Docs.

2.2 Data Cleaning and Exploration

2.2.1 Dataset Description

The data used in this project was provided directly by STUK and consists of three linked datasets, each capturing a different aspect of the radioactivity measurement process. All samples represent environmental materials collected from sites near Finnish nuclear power plants and additional monitoring stations such as Hyytiälä. Sample types are diverse and include fish, lichen, gravel, water, milk, and air, among others.

Dataset 0 contains basic sample metadata, including the sample class (e.g., `air`, `fish`, `milk`), pretreatment type describing the laboratory actions performed on the sample, and the sample group which provides a more specific categorisation within each sample class.

Dataset 1 records physical and preprocessing properties of each sample such as net weight, density, geometry, container dimensions, and vacuum status, as well as binary flags indicating which preprocessing steps were applied - for example drying, ashing, vaporisation, sieving, or the addition of a carrier substance.

Dataset 3 contains the radioactive measurement results for each sample including the detected nuclide (e.g., Cs-137, K-40, Pb-210), the quantified activity value or minimum detectable activity (MDA) where the nuclide was below the detection limit, the measurement unit, and the associated measurement uncertainty expressed as a percentage.

2.2.2 Preprocessing and Data Cleaning

Data cleaning was performed independently by each team member and results were subsequently reviewed and aligned collectively to ensure consistency.

For Dataset 1, rows were removed according to two rules. Firstly, any row containing only an identifier and vacuum status without any accompanying measurement values was discarded as such entries provide no usable information for analysis. Secondly, rows consisting solely of basic metadata fields with no additional data were treated as placeholder entries and similarly removed.

A systematic check for strict rule violations was also carried out on Dataset 1. This involved verifying physical constraints such as whether the gross weight was less than the tare weight (`gross_less_than_tare`), whether the dry weight exceeded the net weight (`dry_gt_netwet`), whether the dry weight percentage exceeded 100 (`dry_percent_gt_100`), and whether the density was non-positive (`density_nonpositive`). Each of these violation types affected approximately 10 samples which represents roughly 0.05% of the dataset. Notably, no violations of the ash weight exceeding dry weight constraint (`ash_gt_dry`) were found. An analysis of sample processing workflows revealed a variety of distinct preparation pathways (e.g., RAW, DRY, VAP, DRY → VAC, VAP → ASH), with the most common being RAW (approximately 12,996 samples) followed by DRY (approximately 3,437 samples).

The raw Dataset 3 consisted of 98,300 rows across 23,170 unique samples and 383 unique nuclide name strings, in long format with one row per nuclide measurement per sample. Parsing the result column introduced an `is_mda` flag to identify the 37,780 rows (38.4%) where measurements fell below the minimum detectable activity. Nuclide names were standardised from 383 strings to 344 canonical names by resolving capitalisation variants, missing hyphens, and two STUK-confirmed typographic errors (`Ra-266` → `Ra-226`; `Ra-238` → `Ra-228`). Each row was then assigned to one of three categories: genuine nuclide measurements (*clean*, 94,454 rows), Finnish-language derived summary quantities such as `Kokonaisalfa` and `Viitteellinen annos` (*derived_quantity*, 3,803 rows), and bare element symbols without mass numbers (*garbage*, 43 rows). The 249 unique unit strings were consolidated into canonical forms and converted to consistent magnitude, including mBq-to-Bq conversions, Bq/g-to-Bq/kg scaling, and tritium unit conversion. U-238 and Th-232 measurements in mass concentration units were flagged separately as they represent a different physical quantity from activity measurements.

Rows were removed following direct confirmation from STUK through a structured question-and-answer process. The largest category of removals was derived quantities deemed unsuitable for nuclide-level analysis (1,211 rows), followed by negative values identified as software artefacts (54 rows), garbage nuclide names (43 rows), rows with a missing unit denominator (23 rows), a confirmed test sample (13 rows), two sets of physically impossible values - Sr-89 at 10^{53}–10^{55} Bq/l in four samples and Bi-214/Pb-214 at 10^{33} Bq/m^3 in two samples, an entry error (8 rows), and samples with a `NULL` sampleClass (4 rows). In total, 1,364 rows were removed, producing a cleaned dataset of 96,936 rows across 23,161 samples.

2.3 Data Exploration, Feature Selection and Clustering

2.3.1 Purpose of Clustering

Clustering was employed as an exploratory technique to identify structure and differences within the samples before supervised anomaly detection was attempted. The goal was to visualise the data in reduced-dimensional space, detect natural groupings, and identify potential outliers for validation. Detected outliers were submitted to STUK, whose domain experts reviewed the flagged samples manually to confirm whether they constituted genuine anomalies.

2.3.2 Clustering Methods

Nine clustering algorithms were evaluated to identify the most suitable approach for the data. Methods were divided among team members, each of whom applied their assigned algorithms to the same preprocessed version of Dataset 1 so that results could be compared on equal footing.

The algorithms spanned three broad families: partition-based methods (K-Means, which assigns each point to the nearest centroid, and Spectral Clustering, which recovers complex-shaped clusters via a similarity graph), density-based methods (DBSCAN, which labels isolated points as noise; OPTICS, which extends DBSCAN to varying-density clusters; and HDBSCAN, which further improves robustness to density variation), and distribution-based methods (GMM, which models the data as overlapping Gaussians with soft cluster memberships). Additionally, Agglomerative Clustering merged the two closest clusters bottom-up in a hierarchical fashion, Mean Shift moved each point toward the local density maximum allowing cluster boundaries to emerge naturally, and an Autoencoder approach first learned a compressed latent representation of the data before K-Means was applied in the latent space.

2.3.3 Results and Insights

For Dataset 1, the autoencoder-based clustering and HDBSCAN yielded the best results. A notable finding common to several density-based and mixture methods was that clusters organised according to sample processing workflow rather than sample type: samples of the same physical class that had undergone different preparation procedures (e.g., RAW versus DRY $\rightarrow$ VAP $\rightarrow$ ASH) were assigned to different clusters. This demonstrates that processing workflow is a stronger discriminating signal in Dataset 1 than sample class alone.

Comparing the autoencoder K-Means method against Mean Shift, the former achieved a higher silhouette score alongside substantially higher Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) values, indicating that the latent features learned by the autoencoder capture the underlying data structure more effectively and produce more clearly separated clusters.

Exploratory analysis of Dataset 1 revealed that the 20,095 pre-processed samples span a wide variety of preparation workflows, with `RAW` being the most common (10,937 samples), followed by `DRY` (3,437), `VAP` $\rightarrow$ `CARRIER` (1,508), and `DRY` $\rightarrow$ `VAC` (1,468). GMM was chosen for clustering due to its soft probabilistic assignment, with n=4 components selected based on the BIC elbow and domain interpretability. The four clusters separated clearly in LDA space (LD1: 67.5%, LD2: 21.8%), though the overall silhouette score of 0.256 reflects the inherently gradual boundaries between sample types in preprocessing data.

Cross-validation with Dataset 0 confirmed that each cluster corresponds to a physically interpretable sample type: Cluster 1 (n=5,703) was 99.6% C-1 air samples, Cluster 2 (n=4,028) captured vaporised liquid samples, Cluster 0 (n=7,313) grouped dried solids, and Cluster 3 (n=3,051) captured fresh biological samples. A notable finding was that C-4 water split across clusters 2 and 3, suggesting two distinct preprocessing workflows for water samples. Further cross-checking with derived workflow labels showed that specific preprocessing steps are near-exclusive signatures of certain sample types - `DRY` $\rightarrow$ `SIEVE` mapped 100% to C-3 soil and `VAP` $\rightarrow$ `CARRIER` was dominated by C-4 water — indicating that GMM clustering was more effective at recovering preprocessing workflow families than discriminating between individual sample types.

Clustering of Dataset 3 was performed using UMAP followed by HDBSCAN, applied independently within each sampleClass, as different sample types are measured for entirely different nuclide sets making global clustering uninformative. Of the 41 sampleClasses, 25 met the inclusion criteria of at least 50 samples and at least two informative nuclides, covering 14,449 of 22,053 samples (65.6%). The remaining classes were excluded due to insufficient samples or nuclide measurements predominantly below the detection limit.

Across the 25 clustered classes, HDBSCAN identified between 2 and 10 clusters per class, with 718 samples (5.0%) flagged as noise. Silhouette scores ranged from 0.23 to 0.80, with nuclide-rich classes such as C-32 industrial products and C-37 ashes producing the clearest separation. Notable findings include C-38 industrial waste splitting into eight sub-groups with substantially different uranium and thorium decay chain profiles, and a C-3 soil cluster showing Cs-137 at 4.6 times the class median with suppressed natural radionuclide levels, consistent with anthropogenic contamination.

Following a STUK recommendation, the analysis was repeated excluding below-detection measurements, removing 37.4% of rows. C-4 water was excluded entirely as Cs-137 and K-40 are almost entirely below detection in water samples, confirming that water radioactivity in this dataset carries little discriminating information. For most classes, removing below-detection values improved cluster quality, with silhouette scores increasing for C-38 ($0.426 \rightarrow 0.628$), C-3 soil ($0.504 \rightarrow 0.622$), C-5 settling material ($0.326 \rightarrow 0.574$), and C-8 periphyton ($0.229 \rightarrow 0.521$). The one exception was C-32 industrial products ($0.796 \rightarrow 0.625$), where the drop suggests that the pattern of which nuclides fall below detection carries meaningful discriminating information. The no-MDA approach is recommended as the primary result, with C-32 noted as a case where both versions should be consulted.

2.4 Model Selection and Implementation

2.4.1 Candidate Models

Following the clustering phase, the focus shifted to training anomaly detection models capable of assigning a label - *typical*, *atypical but plausible*, or *erroneous/exceptional* - to each incoming sample. Because verified ground truth labels were not available for the full dataset, the problem was framed as unsupervised or semi-supervised anomaly detection. Cluster membership and outlier scores produced during the exploration phase were used as a proxy for labelling, and STUK's manual validation of flagged samples provided partial ground truth for evaluation.

Four different models were benchmarked: Isolation Forest, Local Outlier Factor (LOF), Histogram-Based Outlier Score (HBOS), and One-Class SVM.

Isolation Forest

Isolation Forest was tested as a candidate model for Subtask 1. It constructs an ensemble of random decision trees that recursively partition the feature space; anomalous samples require fewer splits to isolate, yielding a lower anomaly score. Being fully unsupervised, it suited our setting where verified ground truth was unavailable.

Training setup: After the same cleaning applied in the clustering phase, approximately 20 095 rows remained from Dataset 1. Dataset 0 metadata was joined via pseudoid, and the 402 samples previously flagged as anomalies by hierarchical clustering were excluded from the training pool. Twenty-one features (11 primary numeric and categorical fields plus 10 sparse columns such as vaporisation and ashing quantities) were label-encoded, median-imputed (fitted on clean data only), and standardised. A PCA pre-reduction retaining 85% of variance was applied before training. The clean pool was split 80/20; the Isolation Forest was configured with 500 estimators, contamination = 0.02, `max_samples` = 256 and `max_features` = 0.8.

Labelling and evaluation: Samples were labelled *Erroneous/Exceptional* (below the 5th percentile of training scores), *Atypical* (5th–20th percentile), or *Typical* (above the 20th percentile). On a test set of 3 939 held-out clean samples plus all 402 known anomalies ($\sim$ 9% prevalence), the model achieved an anomaly-class precision of 0.44, recall of 0.33, F1 of 0.38, and an overall ROC-AUC of 0.75. Only 134 of the 402 known anomalies (33%) were correctly flagged as Erroneous/Exceptional; the remaining 268 blended into the Typical or Atypical categories, indicating that Isolation Forest captured the most extreme outliers but struggled with subtler deviations.

2.4.2 Selected Model and Rationale

Local Outlier Factor (LOF) was selected as the primary anomaly detection model on the basis of achieving the best anomaly detection accuracy among the candidates.

2.4.3 Performance

For Dataset 1 (Appendix B.1), the LOF model achieved strong overall classification performance. The model obtained an overall accuracy of 93.16%. For the normal class, the precision was 96.62%, recall was 95.18% and the F1-score was 96.21%. For the anomaly class, the precision was 62.07%, recall was 67.16% and the F1-score was 64.52%.

For Dataset 3 (Appendix B.1), anomaly detection proved to be more challenging mainly because of the high variability of radioactive measurement data across different sample classes and nuclide combinations. The LOF model achieved overall accuracy of 92.97%. For the normal class, the precision was 94.12%, recall was 98.16% and the F1-score was 96.10%. For the anomaly class, the precision was 79.76%, recall was 54.14% and the F1-score was 64.50%.

3. Deliverables

3.1 Results

The project produced two prototype classification systems, one for each subtask defined by STUK, together with the Jupyter notebooks that document the full analytical pipeline from data cleaning to model evaluation.

The primary deliverable for both subtasks is the Jupyter notebook implementation. Because STUK intends to integrate the classification logic into their existing internal tooling rather than adopt a separate application, self-contained notebooks provide the most practical handover artefact as the code can be inspected, modified, and merged into STUK's current workflows without the legal and operational overhead that a standalone third-party application would require. The Streamlit web applications described in the following sections were developed in parallel as working demonstrations and for interactive validation, but they are supplementary to the notebook deliverables.

3.2 Pipeline

3.2.1 Subtask 1 - Preprocessing Pipeline

The Subtask 1 pipeline trains a Local Outlier Factor model from two reference CSV files: a set of confirmed normal samples (`dataset1_final.csv`) and a set of identified outliers (`dataset1_outliers.csv`). After dropping the identifier column, shared numeric features are median-imputed, categorical features are mode-imputed, and all features are one-hot encoded and scaled with a `RobustScaler`. The LOF model is fitted with `n_neighbors=12` and `contamination=0.04` in novelty mode on 80% of the normal data. The *atypical* threshold is set at the 90th percentile of normal scores; the *erroneous* threshold is chosen from percentiles 80–99 by maximising F1 on a held-out validation set combining the remaining normal samples and the outlier set.

At inference time, a sample is preprocessed using the fitted transformers and assigned one of three labels based on its LOF score relative to the two thresholds. The output additionally includes a percentile rank and a plain-language explanation identifying any numeric features with a robust z-score above 2.0.

3.2.2 Subtask 2 - Measurement Results Pipeline

The Subtask-2 prototype implementation uses Local Outlier Factor (LOF) model trained on preprocessed radionuclide measurement values. The raw measurement data is transformed into a sample level feature matrix where each sample contains the measured nuclide activity values as features. Preprocessing includes handling missing values, logarithmic scaling, robust scaling and dimensionality reduction using Principal Component Analysis (PCA).

The anomaly list that was used for the prototype testing and threshold calibration was derived from the clustering phase of subtask 2 using UMAP+HDBSCAN clustering results. After the training, pipeline can classify new input files automatically and produce a CSV output containing the pseudoid, predicted class, anomaly score and the nuclide values, which contributed the most to the anomaly score.

The final prototype was delivered as a Jupyter notebook so that the classification logic can be implemented into STUK's existing workflow and be modified further if needed.

3.3 User Interface Implementation

Two Streamlit applications were built as interactive demonstrations alongside the pipeline code. As noted earlier, the primary handover deliverable is the Jupyter notebook implementation, which STUK can integrate directly into their existing tooling without the legal and procurement overhead of adopting a standalone third-party application.

Both applications share the same general structure: a sidebar for mode selection and configuration, a *Classifier* tab supporting both single-sample entry and CSV batch upload, and a *History* tab that records every run as a JSON file for later review and download. Results are displayed on colour-coded cards (green / amber / red for Typical / Atypical / Erroneous) with supporting charts and per-feature diagnostics.

The Subtask 1 app adds configurable data quality thresholds for missing value ratios, triggering PASS, REVIEW, or BLOCKER statuses before classification proceeds. The Subtask 2 app additionally exposes the four nuclide selection hyperparameters in the sidebar and allows the engine to be retrained in-app without restarting the server. Samples of the user-interfaces are shown in Appendix A.

4. Challenges and Future Direction

The most significant obstacle was the absence of verified ground truth labels, making it impossible to evaluate anomaly detection performance in a conventional supervised sense. Outliers were submitted to STUK for manual validation, but this introduced delays of up to one month before confirmed labels were available for model calibration. Selecting appropriate clustering methods added further difficulty, as nine algorithms were evaluated without ground truth to guide comparison, adding approximately two weeks to the project timeline.

Looking ahead, the most impactful improvement would be to use a manually validated anomaly list as ground truth for proper model calibration and evaluation. Additional machine learning models should be explored for Dataset 3, and a user interface for the classification pipeline would allow STUK analysts to submit new samples and receive results without running the underlying code directly.

5. Conclusion

This project demonstrated that anomaly detection methods can be effectively applied to STUK's environmental radioactivity measurement data to reduce the burden of manual inspection. Through extensive preprocessing, validation, clustering, and model development, the project resulted in a complete workflow capable of identifying atypical and erroneous samples from large datasets.

The exploratory analysis and clustering revealed meaningful structure within radioactive measurement results. In particular, the analyses showed that preprocessing workflows often form stronger grouping signals than sample type alone, and that radioactive measurement patterns vary significantly across sample classes. These findings provided valuable insights from the data and helped guide the selection of suitable anomaly detection methods.

Among the evaluated models, Local Outlier Factor (LOF) achieved the best overall performance and was therefore selected as the main anomaly detection approach. The resulting models were able to classify samples into typical, atypical, and erroneous categories with high overall accuracy.

In addition to the analytical results, the project delivered practical tools for STUK's future use. The final deliverables include Jupyter notebooks suitable for integration into existing workflows, as well as Streamlit-based user interfaces that allow for interactive analysis and validation. Together, these outputs demonstrate practical value of automated anomaly detection in environmental radioactivity monitoring.

Although the models already provide strong baseline performance, several opportunities for future improvement remain. These include incorporating larger quantities of validated anomalies, exploring other anomaly detection methods, and extending the approach to additional datasets and monitoring contexts. Nevertheless, the project successfully established a scalable foundation for automated anomaly detection at STUK and showed how machine learning can support more efficient and reliable environmental radiation monitoring in the future.

Appendix A. Prototype Screenshots

The following screenshots show the Streamlit prototype interfaces for Subtask 1 and Subtask 2.

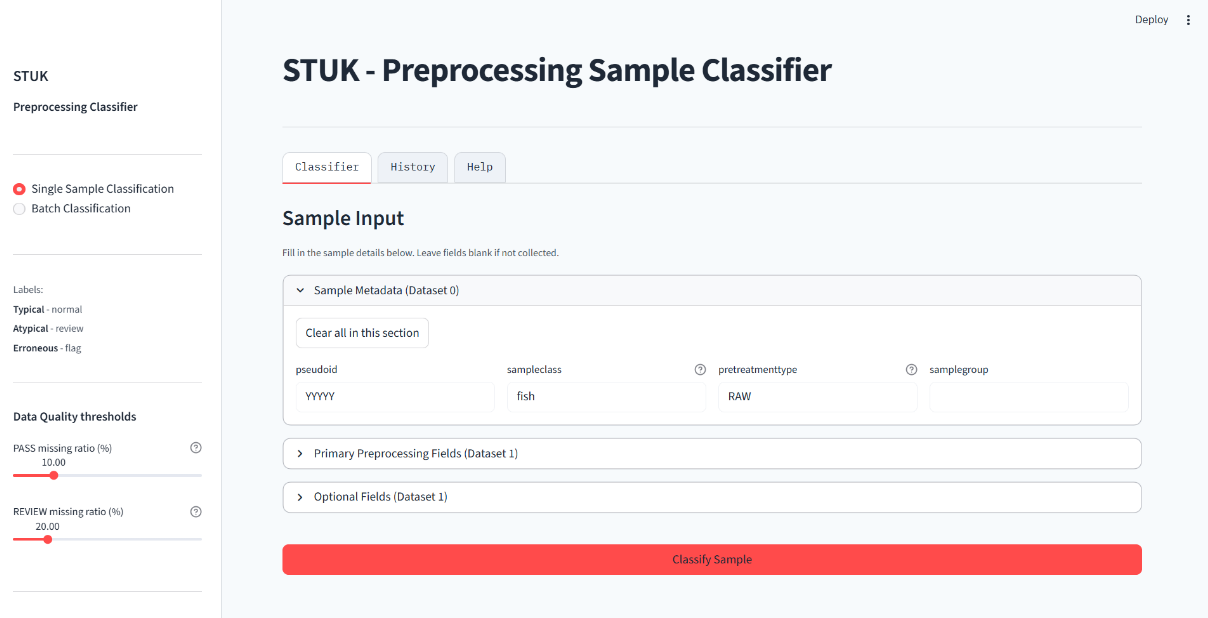

Figure A.1: Subtask 1 Prototype

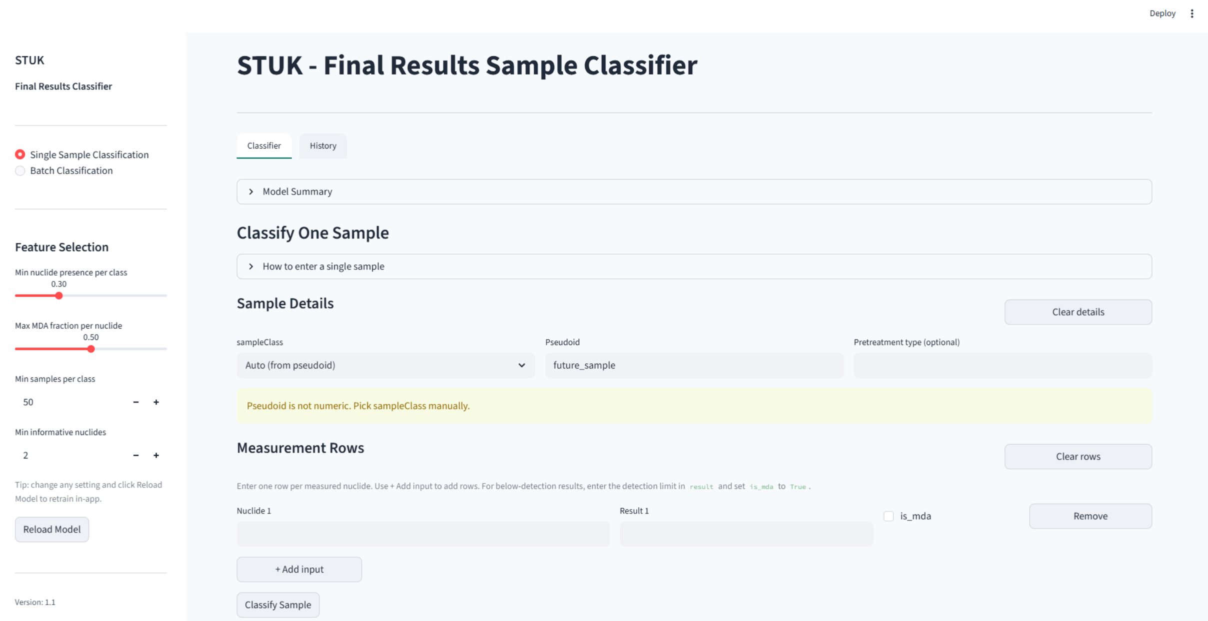

Figure A.2: Subtask 2 Prototype

Appendix B. LOF Confusion Matrix

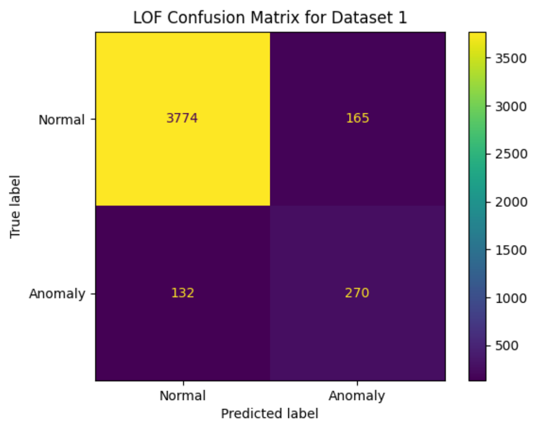

Figure B.1: LOF confusion matrix for Dataset 1

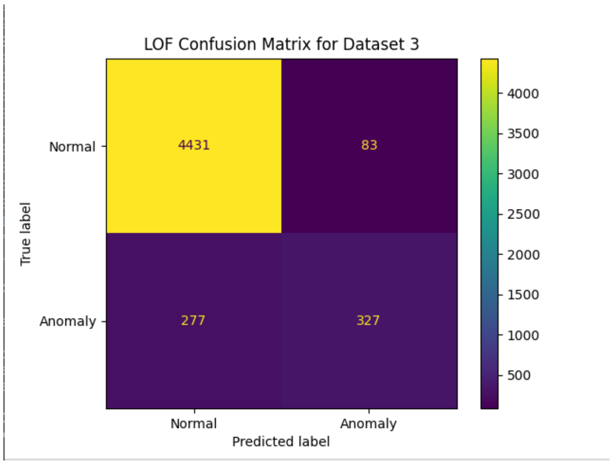

Figure B.2: LOF confusion matrix for Dataset 3

Appendix C. Clusters of SampleClass

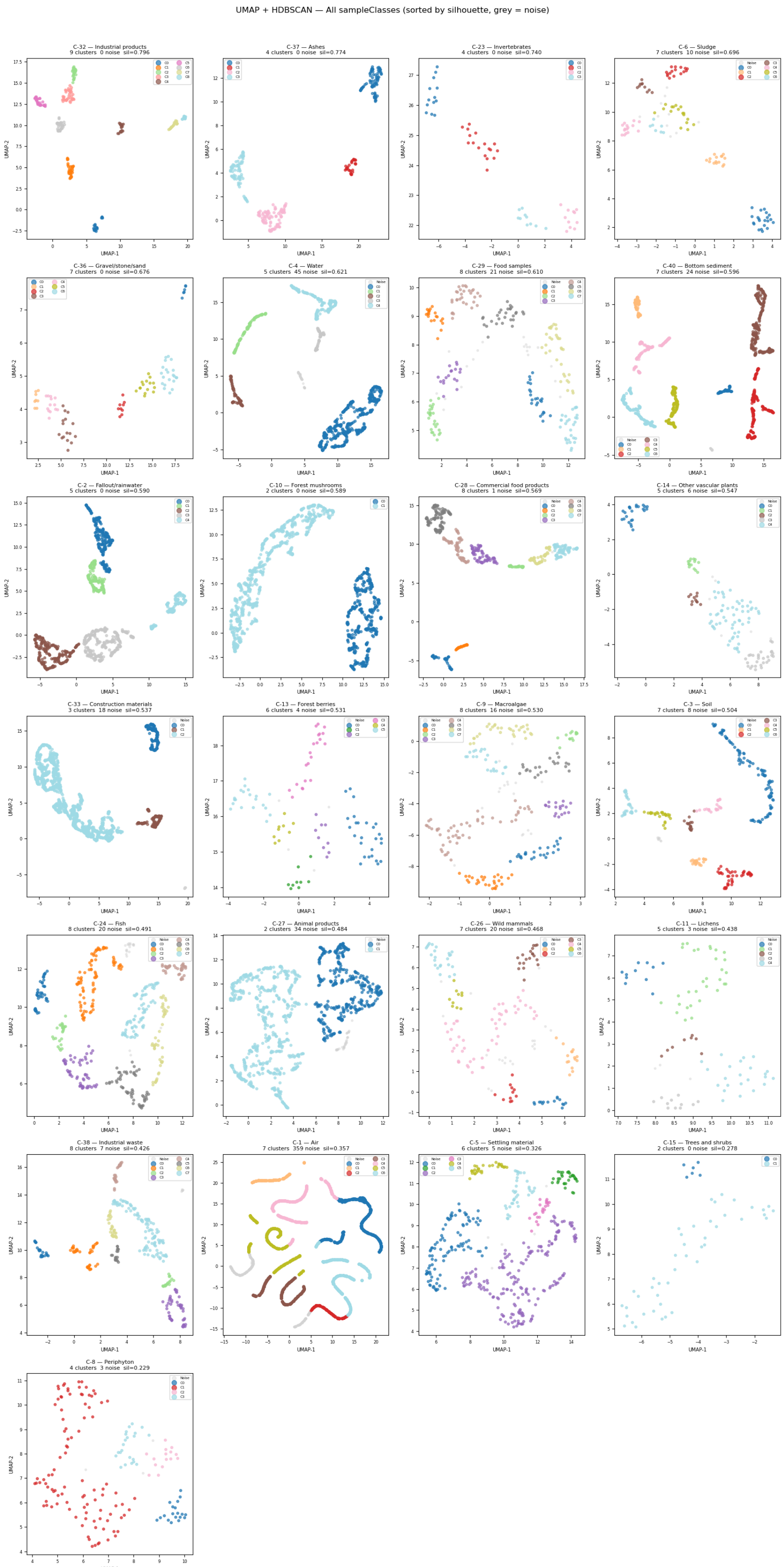

Figure C.1: Clusters of samples for Dataset 3

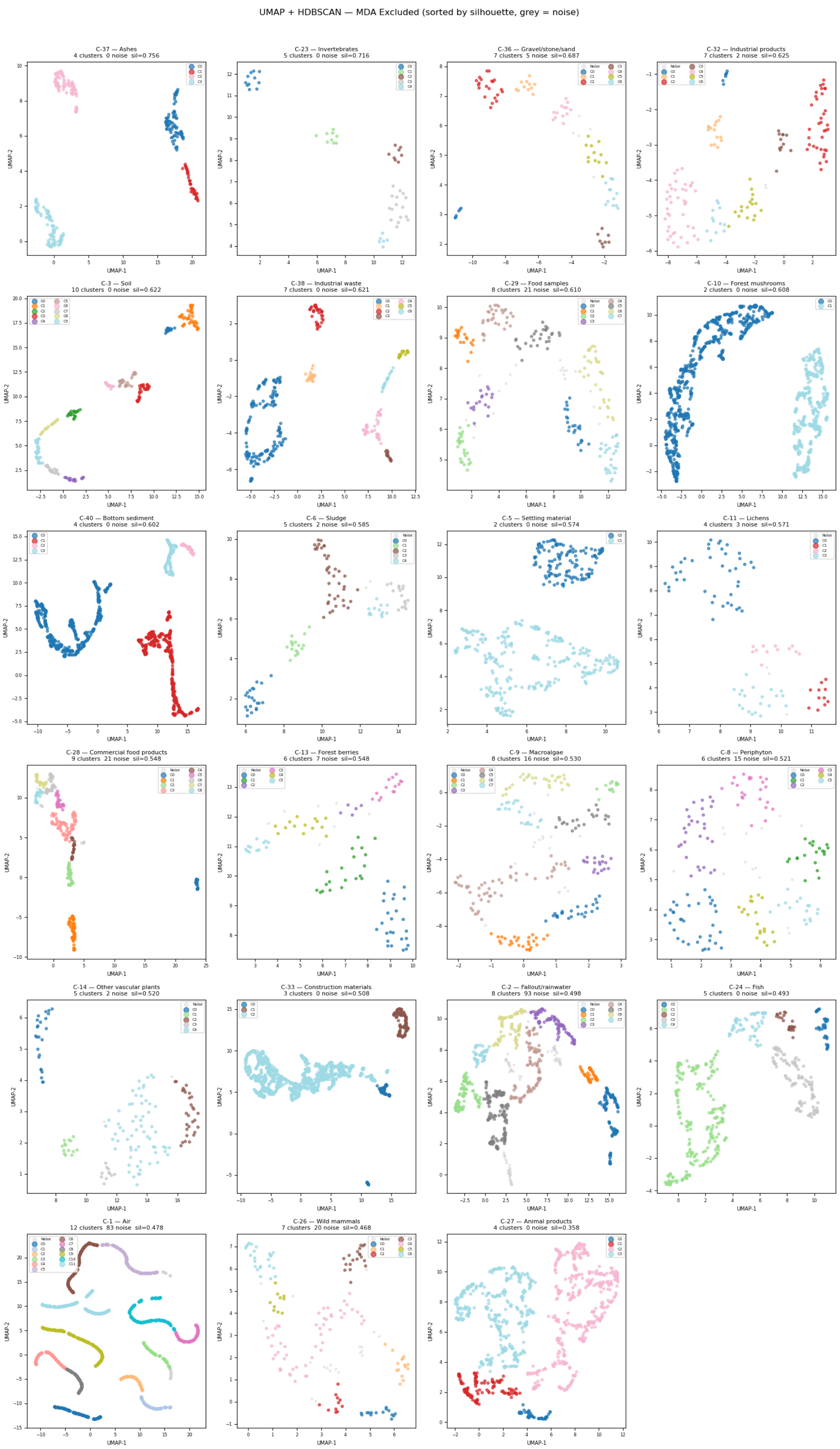

Figure C.2: Clusters of samples for Dataset 3 excluding MDA